Virtual Memory – MCQ Practice

20 Multiple-Choice Questions | Medium Difficulty

Instructions: Select the single best answer for each question.

1. What is the primary purpose of virtual memory in an operating system?

A. To increase physical RAM without hardware changes
B. To allow processes to execute without their entire address space being in physical memory
C. To eliminate secondary storage
D. To make every process use the same physical addresses

2. A system has 4 GB of RAM, but running processes require more than 4 GB of virtual memory. Why can they still execute?

A. The OS increases RAM frequency
B. The CPU executes everything directly from disk
C. Only required portions need to be resident in physical memory at a given time
D. Virtual memory removes the need for physical memory

3. Which statement correctly describes demand paging?

A. All pages are loaded before execution
B. Pages are loaded into memory when needed
C. Pages are loaded only when a process terminates
D. Pages are permanently stored in the TLB

4. A process accesses a valid virtual page that is not currently in physical memory. What normally happens?

A. The CPU immediately terminates the process
B. The OS handles a page fault and arranges for the page to become available
C. The page table is permanently deleted
D. Execution continues without OS intervention

5. Which situation can cause a page fault even when the virtual address belongs to a valid process address space?

A. The required page is not loaded into physical memory
B. The CPU cache contains the data
C. The TLB contains a valid translation
D. The page is present and accessible in a physical frame

6. A parent creates a child using fork() with Copy-on-Write (COW). What happens initially to shared memory pages?

A. Every page is immediately copied
B. Both processes may share physical pages until one modifies a shared page
C. The child cannot access the parent's memory contents
D. All shared pages are immediately written to disk

7. Under COW, a child tries to modify a shared page marked read-only for COW. What typically happens?

A. Both processes are terminated
B. The write is ignored
C. The OS handles the protection fault, creates a private copy when needed, and allows the write
D. FIFO replacement is automatically performed

8. When does an OS generally need page replacement?

A. Whenever a process creates a thread
B. Whenever the TLB records a hit
C. When a needed page must be loaded and no free frame is available
D. Every time a process makes a system call

9. A system has three initially empty frames. Reference string: 2, 3, 1, 2, 4. Using FIFO, which page is evicted when page 4 is accessed?

A. Page 2
B. Page 3
C. Page 1
D. No page is evicted

10. Which characteristic is associated with FIFO page replacement?

A. Replaces the least recently used page
B. Replaces the page that entered memory earliest
C. Replaces the page used farthest in the future
D. Always produces the fewest page faults

11. A system uses LRU with three initially empty frames. Reference string: 4, 2, 1, 4, 3. Which page is evicted when page 3 is accessed?

A. Page 4
B. Page 2
C. Page 1
D. Page 3

12. Why can LRU perform better than FIFO for some reference patterns?

A. LRU considers recent usage, while FIFO considers arrival order
B. LRU knows future references
C. LRU does not generate page faults
D. LRU requires no usage tracking

13. What is the defining principle of Optimal page replacement?

A. Replace the page that entered memory first
B. Replace the most recently accessed page
C. Replace the page whose next use is farthest in the future
D. Replace the page with the smallest number

14. Why is Optimal page replacement not generally implementable as a practical online algorithm?

A. It requires every page to be in RAM
B. It requires knowledge of future page references
C. It always causes more faults than FIFO
D. It cannot be used with virtual memory

15. For a fixed reference string and the same number of frames, which statement is correct?

A. FIFO always has fewer faults than LRU
B. LRU always has fewer faults than Optimal
C. Optimal yields the minimum possible page faults under the stated model
D. All algorithms have the same number of faults

16. What does frame allocation determine?

A. The number of CPU cores per process
B. The number of physical frames assigned to each process
C. The number of files each process can open
D. The number of instructions in a process

17. Equal allocation distributes 60 available frames among 5 processes. How many frames does each receive?

A. 10
B. 12
C. 15
D. 20

18. A process has too few frames for its actively used pages. What is a likely consequence?

A. Its page-fault rate may increase
B. Its page-fault rate must become zero
C. Its virtual address space shrinks automatically
D. Its page table is no longer needed

19. What best describes thrashing?

A. The CPU spends most time executing useful instructions
B. The OS spends excessive time handling page faults and moving pages between memory and storage
C. The TLB stores too many valid translations
D. The system has more free frames than needed

20. A system has sharply increasing page faults and falling CPU utilization as more processes enter memory. What is most likely to reduce thrashing?

A. Admit even more processes
B. Reduce frames available to each process
C. Reduce multiprogramming so active processes have more memory
D. Disable page replacement

Answer Key

Question	Answer	Question	Answer
Q1	B	Q11	B
Q2	C	Q12	A
Q3	B	Q13	C
Q4	B	Q14	B
Q5	A	Q15	C
Q6	B	Q16	B
Q7	C	Q17	B
Q8	C	Q18	A
Q9	A	Q19	B
Q10	B	Q20	C